DT124A - USART and Timer-based Communication using Arduino

Introduction

This is a revised version of the lab for 2025. The lab requires no written report and focuses instead on in-lab demonstration and discussion.

Good luck, R

Purpose

The purpose of this lab is:

- Learn how to set up and use the USART hardware for serial communication.
- Use polling to receive input from a PC via serial connection.
- Implement basic delay functionality using a timer peripheral.
- Practice multi-device communication between Arduinos. (For higher grade)

Equipment

- Laptop computer (students' own) with Microchip studio IDE (or equivalent IDE that supports assembly language development for AVR/Atmel processors).
- Arduino Uno (or similar microcontroller) and AVR programmer
- Red LED
- Resistors (1 x 220Ω).
- Tools → Data Visualizer for the serial communication between PC and MCU. Or use the Serial Monitor in Arduino IDE.
- You can use ch19. USART0 in the Datasheet (page 143–163) as a help for the USART Communication.

Preparation

Before coming to the lab, make sure you:

1. Read chapter 19 of the ATmega328P datasheet (pages 143–163) on USART0.
2. Review Timer and delay functionality (e.g., Timer0 or Timer1).
3. Understand polling vs interrupt-based I/O.

Grading

Grade 3 - Passing Grade

- Task 1 completed and demonstrated.
- Able to explain code structure and logic.
- Willingness to make small changes upon request.

Grade 4

- Task 1 and Task 2 completed and demonstrated.
- Able to answer deeper questions about USART configuration.
- Can make moderate modifications to functionality.

Grade 5

- All three tasks completed and working.
- Can explain and justify timer and USART setup.
- Able to make modifications and demonstrate initiative.

1 Task 1 – UART Blink using Polling

Write a program where you make the LED blink on and off every second if you send a '1' from the PC to the Arduino; if you send a '0', the same LED should remain off.

Requirement:

- You should use the F_CPU of $16000000UL$.
- You should use BAUD of 9600.
- You should use an 8-bit frame, 1 stop bit, no-parity in the `UCSR0C` register.
- You should poll the `RXC0` bit to find out whether you have received data.

- You should use a delay function that is implemented by a Timer on the Arduino.

Use the Data Visualizer (Tools → Data Visualizer) or Serial Monitor in Arduino IDE to transmit data from the PC to your MCU.

2 Task 2 - USART Communication

Write a program that takes the input from the serial monitor and then when an end of line character is reached the Arduino then sends back the input in reverse.

Requirement:

- You should use the serial monitor to send the message
- When an "newline" character is reached the Arduino should send the message back reversed.
- If the "newline" character cannot be made to work you can choose you own EOL.

Use the Data Visualizer (Tools → Data Visualizer) or Serial Monitor in Arduino IDE to transmit data from the PC to your MCU.

Remember that there exists buffer size limits.

3 Task 3 – UART Communication Between Two Arduinos

Connect your PC to an Arduino (called A1) that is connected to another Arduino (called A2) that then is connected to a LED-light. If a command '0' is sent to A1 it relays it to A2 which blinks a LED with 1 s delay. If the command '1' is sent, it blinks the LED with 4 s delay.

Requirements:

- Two Arduino units connected needs to be used
- The Arduino connected to the PC cannot be the one blinking the LED